Adjoint Functors - A Brief Introduction

Ryan Thompson

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Te Herenga Waka - Victoria University of Wellington

Motivation

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Conversely, given any set S, it is possible to construct the "free" group on S, which in some sense is the most general group one can associate with S.

A basic example of a free group is the integers, which is the unique (up to isomorphism) free group with one generator (1)!.

Motivation

In algebra there exists the notion of a Galois connection, which is a pair of functions between preordered sets $f : A \rightleftarrows B : g$ such that

$$f(a) \leq b \iff a \leq g(b)$$

for all $a \in A$ and $b \in B$.

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for all $a \in A$ and $b \in B$.

At first glance, this seems like an entirely unrelated concept to the free / faithful relationship - but we will see that these are actually the same category theoretic idea at work.

Prelims

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Definition
A functor between two categories C and D is an assignment of an object $F(c) \in C$ for each object $c \in C$, and an arrow $F(f) : Fc \to Fd$ in D for every arrow $f : c \to d$ in C. We require that $F(id_c) = id_{Fc}$, and that the following diagram commutes.

$$\begin{tikzcd} c \arrow[r, "f"] \arrow[d, "F"] & d \arrow[d, "F"] \\ Fc \arrow[r, "Ff"] & Fd \end{tikzcd}$$

A functor of this form is called covariant, to emphasize the fact that it preserves the direction of arrows.

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If c, d are objects of a category C, we denote by $\hom(c, d)$ the collection of arrows from c to d. We will assume all of our categories are small, such that $\hom(c, d)$ is a set (as opposed to a class).

Definition

Let C, D be categories, and $F, G : C \to D$ two functors. A natural transformation between F and G, denoted by $F \Rightarrow G$, consists of a collection of arrows $\alpha_c : Fc \to Gc$ for every object $c \in C$, such that the following naturality diagram commutes for any arrow $f : c \to d$ in C:

$$\begin{tikzcd}
Fc \arrow[r, "\alpha_c"] \arrow[d, "Ff"] & Gc \arrow[d, "Gf"] \\
Fd \arrow[r, "\alpha_d"] & Gd
\end{tikzcd}$$

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We say that a natural transformation is a natural isomorphism if each of the arrows α_c above is an isomorphism (i.e. has an inverse).

Example
Consider the category Set of sets, and the functor $P : \mathrm{Set} \to \mathrm{Set}$ that maps a set A to its powerset $P(A)$.

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There is a natural transformation $\mathrm{id}_{\mathrm{Set}} \Rightarrow P$, such that each component $\alpha_A : A \to P(A)$ maps an element $a \in A$ to the singleton $\{a\} \in P(A)$.

Prelims

Definition
Let C be a category, and $c \in C$ an object. The covariant hom functor, denoted $\hom(c, -)$, maps an object $d \in C$ to the set $\hom(c, d)$, and an arrow $f : x \to y$ to the post-composition function $f_* : \hom(c, x) \to \hom(c, y)$ defined by $f_*(g) = f \circ g$.

$$\begin{tikzcd} & c \arrow[dl, "g"'] \arrow[dr] & \\ x \arrow[rr, "f"] & & y \end{tikzcd} \quad \mapsto \quad \begin{tikzcd} & c \arrow[dl, "g"'] \arrow[dr] & \\ x \arrow[rr, "f"] & & y \end{tikzcd}$$

Prelims

Definition
The contravariant hom functor, denoted $\hom(-, c)$, maps an object $d \in C$ to the set $\hom(d, c)$, and an arrow $f : x \to y$ to the pre-composition function $f^* : \hom(y, c) \to \hom(x, c)$, defined by $f^*(g) = g \circ f$.

$$
\begin{tikzcd}
x \arrow[rr, "f"] \arrow[dr] & & y \arrow[dl, "g"] \\
& c &
\end{tikzcd}
\quad \mapsto \quad
\begin{tikzcd}
x \arrow[rr, "f"] \arrow[dr] & & y \arrow[dl, "g"] \\
& c &
\end{tikzcd}
$$

Adjoints

First Definition

The first way we might define adjoint functors establishes a natural isomorphism between particular hom-sets.

Definition
Let C, D be categories, and $F : C \rightleftarrows D : G$ functors, such that there is a natural isomorphism

$$\hom(Fc, d) \cong \hom(c, Gd)$$

for each $c \in C$ and $d \in D$. Then we say that F, G comprise an adjunction between C and D. In particular, we say F is left-adjoint to G, and G is right-adjoint to F (this is because F appears on the left-side of the hom-functor in the equation above, and G appears on the right!)

We write $F \dashv G$ to indicate that F is left-adjoint to G.

First Definition

The equation above might remind you of the following from functional analysis...

Definition
Let H be a Hilbert space, and A a bounded linear operator. The adjoint of A, denoted A^*, is the bounded linear operator satisfying

$$\langle Ax, y\rangle = \langle x, A^* y\rangle,$$

for all $x, y \in H$.

This is where the name "adjoint" in "adjoint functor" came from. However, the A and A^* here aren't "adjoint functors" in any natural sense.

First Definition

This is stating that there is a one-to-one correspondence between arrows $f : Fc \to d$ and $g : c \to Gd$ that is natural.

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Sometimes, the pair of corresponding morphisms given by this isomorphism are denoted $f^\sharp$ and $f^\flat$.

$$
\begin{tikzcd}
Fc \arrow[d, "f^\sharp"] & \leftrightarrow & c \arrow[d, "f^\flat"] \\
d & & Gd
\end{tikzcd}
$$

For the second definition of an adjunction, we need to know about whiskering!

This is a way to "horizontally" compose a functor with a natural transformation to acquire another natural transformation. More explicitly, let $F : C \to D$ be a functor, and $G, H : D \to E$ be functors with a natural transformation $\alpha : G \to H$ as below.

$$\begin{tikzcd} C \arrow[r, "F"] & D \arrow[r, bend left, "G"] \arrow[r, bend right, "H"'] \arrow[r, phantom, "\Downarrow\alpha"] & E \end{tikzcd}$$

Interlude - Whiskering

We can then define a natural transformation $\alpha F : GF \Rightarrow HF$ with component α_{Fc} at $c \in C$.

$$
\begin{tikzcd}
 & & GFc \arrow[dd, Rightarrow, "\alpha_{Fc}"] \\
c \arrow[r, "F"'] & Fc \arrow[ur, "G"'] \arrow[dr, "H"'] & \\
 & & HFc
\end{tikzcd}
$$

Definition
Let C, D be categories, and $F : C \leftrightarrows D : G$ functors, such that there exist natural transformations

$$\eta : 1_C \Rightarrow GF, \quad \epsilon : FG \Rightarrow 1_D$$

such that the following diagrams commute, where $F\eta$ and ϵF are whiskered natural transformations (similarly for ηG and $G\epsilon$).

$$\begin{tikzcd} F \arrow[r, Rightarrow, "F\eta"] \arrow[dr, Rightarrow, "\mathrm{id}_F"'] & FGF \arrow[d, Rightarrow, "\epsilon F"] \\ & F \end{tikzcd} \qquad \begin{tikzcd} G \arrow[r, Rightarrow, "\eta G"] \arrow[dr, Rightarrow, "\mathrm{id}_G"'] & GFG \arrow[d, Rightarrow, "G\epsilon"] \\ & G \end{tikzcd}$$

For any object $c \in C$, we call the arrow $\eta_c : c \to GFc$ the unit of the adjunction, and for any $d \in D$ we call the arrow $\epsilon_d : FGd \to d$ the counit of the adjunction.

Examples

Example - Free $\dashv$ Forgetful

There are a large class of functors that are "forgetful" in that they "forget" some of the structure or information associated with an object.

e.g. $U : \mathrm{Ab} \to \mathrm{Set}$ that maps an abelian group to its underlying set. To any set A, we can associate a free abelian group that is generated by A modulo the group axioms (i.e. formal $\mathbb{Z}$-linear combinations of elements of A). Let $F : \mathrm{Set} \to \mathrm{Ab}$ be the free abelian group functor. Then the pair (U, F) is a free/forgetful adjunction, and U is left-adjoint to F!

Example - Free $\dashv$ Forgetful

We can understand this adjunction by describing its unit and counit. Notice that the functor $UF : \mathrm{Set} \to \mathrm{Set}$ takes a set A to the set of formal $\mathbb{Z}$-linear combinations of elements of A (emphasis on set).

The component of the unit $\mathrm{id}_{\mathrm{Set}} \Rightarrow GF$ at $A \in \mathrm{Set}$ maps an element $a \in A$ to the trivial combination $a \in UFA$.

Notice that the functor $FU : \mathrm{Ab} \to \mathrm{Ab}$ takes an abelian group G to the abelian group of formal $\mathbb{Z}$-linear combinations of elements of the underlying set of G.

The component of the counit $GF \Rightarrow \mathrm{id}_{\mathrm{Ab}}$ takes a formal sum $\sum_{i=0}^{n} a_i \alpha_i$ of elements of an abelian group G to its evaluation as an element of G.

Interlude - Poset Categories

A poset (short for partially ordered set) is a pair $(A, \leq)$ where A is a set, and $\leq$ is a relation on A that is reflexive, antisymmetric and transitive.

Classic example - the powerset of a set X with set inclusion $\subseteq$.

Any poset $(A, \leq)$ can be considered as a category, where the objects are elements of A, and there is an arrow $x \to y$ if and only if $x \leq y$.

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What does an adjunction look like?

Example - Galois Connection

Let C and D be poset categories. An adjunction between C and D is a pair of functors $F : C \leftrightarrows D : G$ such that

$$Fx \leq y \iff x \leq Gy \quad \text{for all } x \in C, y \in D.$$

In this context, we often refer to F as the lower (Galois) adjoint and to G as the upper (Galois) adjoint (for obvious reasons)! Furthermore, the pair $F : C \leftrightarrows D : G$ is called a Galois connection.

Example - Galois Connection

Let A, B be sets with powersets $P(A)$ and $P(B)$. If $f : A \to B$ is a function, there are two natural ways to induce a function between $P(A)$ and $P(B)$...

1: the direct image function f^* takes a subset $C \subseteq A$ to the direct image $f(C) \subseteq B$;

2: the preimage function f_* takes a subset $C \subseteq B$ to the preimage $f^{-1}(C) \subseteq A$.

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2: the preimage function f_* takes a subset $C \subseteq B$ to the preimage $f^{-1}(C) \subseteq A$.

Thus, the pair $f^* : P(A) \leftrightarrows P(B) : f_*$ form an adjunction! This follows from the fact that if $A' \subseteq A$ and $B' \subseteq B$, then $f(A') \subseteq B'$ if and only if $A' \subseteq f^{-1}(B')$.

This establishes a bijection $\hom(f^*(A'), B') \cong \hom(A', f_*(B'))$ (notice that in a poset category, each hom-set contains either 0 or exactly 1 arrow!)

Monads

Monads

A monad on C is a monoid in the category of endomorphisms on C

We will first see the classic definition, and then an alternative definition via adjunctions.

Definition

A monad on a category C consists of an endofunctor $T : C \to C$, along with a natural transformation $\eta : 1_C \Rightarrow T$ called the unit, and a natural transformation $\mu : T^2 \Rightarrow T$ such that the following diagrams commute.

$$
\begin{tikzcd}
T^3 \arrow[r, Rightarrow, "T\mu"] \arrow[d, Rightarrow, "\eta T"] & T^2 \arrow[d, Rightarrow, "\mu"] \\
T^2 \arrow[r, Rightarrow, "\mu"] & T
\end{tikzcd}
\qquad
\begin{tikzcd}
T \arrow[r, Rightarrow, "\eta T"] \arrow[dr, Rightarrow, "\mathrm{id}_T"'] & T^2 \arrow[d, Rightarrow, "\eta"] & T \arrow[l, Rightarrow, "T\eta"'] \arrow[dl, Rightarrow, "\mathrm{id}_T"] \\
& T &
\end{tikzcd}
$$

The first diagram amounts to saying there are two ways to whisker μ with T, and they give you the same result.

$$\begin{tikzcd}[column sep=huge]
C \arrow[r, "T"] \arrow[rr, bend left=40, "T"] \arrow[rr, phantom, bend left=20, "\mu\Uparrow"] & C \arrow[r, "T"] \arrow[rr, bend right=40, "T"'] \arrow[rr, phantom, bend right=20, "\Downarrow\mu"] & C \arrow[r, "T"] & C
\end{tikzcd}$$

The second diagram amounts to saying there are two ways to whisker η with T, and they yield the same result.

$$\begin{tikzcd}[column sep=huge]
C \arrow[r, "T"] \arrow[rr, bend left=40, "T"] \arrow[rr, phantom, bend left=20, "\eta\Downarrow"] & C \arrow[r, "T"] \arrow[rr, bend right=40, "T"'] \arrow[rr, phantom, bend right=20, "\Uparrow\eta"] & C \arrow[r, "T"] & C
\end{tikzcd}$$

"All these arrows are nice, but it's just a bunch of abstract nonsense!"

Monads from Adjunctions

In her book "Category Theory in Context", Emily Riehl says

"Monads are the shadows cast by adjunctions"

Recall that a monad is an endofunctor with additional structure.

Every adjunction $F : C \leftrightarrows D : G$ gives rise to a monad on C given by the endofunctor GF.

The unit $\mathrm{id}_C \Rightarrow GF$ is given by the unit of the adjunction.

The whiskered counit $G\epsilon F : GFGF \Rightarrow GF$ is the multiplication for the monad.

$$\begin{tikzcd}
 & & C \arrow[dr, "F"] \arrow[d, phantom, "\Downarrow\epsilon"] & & \\
 & D \arrow[ur, "G"] \arrow[rr, "\mathrm{id}_D"] & {} & D \arrow[dr, "G"] & \\
C \arrow[ur, "F"] & & & & C
\end{tikzcd}$$

Monads - Example

Consider the free $\dashv$ forgetful adjunction between the categories Set of sets, and Set_* of pointed sets (a set with a distinguished element).

The forgetful functor $U : \mathrm{Set}_* \to \mathrm{Set}$ forgets that the distinguished point is special, while the free functor $F : \mathrm{Set} \to \mathrm{Set}_*$ maps a set A to the pointed set $A_+ := A \cup \{A\}$.

Monads - Example

The unit of the monad $\eta : \mathrm{id}_C \Rightarrow UF$ is the obvious inclusion $A \hookrightarrow A \cup \{A\}$.

The multiplication component $\mu : UFUF \Rightarrow UF$ at an object

$$(A_+)_+ := (A \cup \{A\}) \cup \{A \cup \{A\}\}$$

behaves as follows:

$$\mu_{(A_+)_+}(a) = \begin{cases} \text{a} & \text{if } a \in A, \\ \{A\} & \text{otherwise.} \end{cases}$$

This monad on Set is called the maybe monad - in certain contexts we want to consider total maps $f : A \to B_+$, where the preimage of the distinguished point of B_+ are considered to be the points in A on which f is "undefined".

Monads - My Favourite Example!

Let C, D be poset categories, and $F : C \leftrightarrows D : G$ an adjunction, such that GF is a monad on C. What does this mean?

It means that $c \leq GFc$, and $GFGFc \leq GFc$ - thus $GFGFc = GFc$ - for all $c \in C$. This is exactly what a closure operator is!

For instance, let C be the poset of open subsets of a topological space X. If F is a monad on C, then for all $U \subseteq X$, we have $U \subseteq FU$, and $F^2U = FU$.

Under this definition, the operation "closure of an open set" in topology is a monad on its poset of open sets! (very cool)